

- There are 7 Questions for 50 marks.
 - Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class. You may use any result from class directly (without re-deriving the proof). Clearly justify any statement you make, even if the problem doesn't explicitly ask you to.
 - Good luck!
- (a) [2] Given $m \times n$ matrices A, B and $\underline{y} \in \mathbb{R}^n$, what is the vector $\underline{x} \in \mathbb{R}^n$ that minimizes $\|A\underline{x} - B\underline{y}\|_2$?
 - (b) [2] Suppose that any column of A is orthogonal to any column of B . How would you give a 'simple' equation (involving matrices A, B , transposes and matrix products) that captures this relationship?
 - (c) [2] In case A, B satisfy the property in (b) above, what is the corresponding answer to (a)?
 - Consider the following sequence of matrices

$$A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad A_n = \begin{pmatrix} A_{n-1} & I \\ I & -A_{n-1} \end{pmatrix} \text{ for } n \geq 2.$$

- (a) [2] What is the size of the matrix A_n ?
 - (b) [2] Is A_n symmetric?
 - (c) [2] What is A_n^2 ? *Hint: Guess a pattern, then prove by induction*
 - (d) [2] What is the rank of A_n ?
- Consider the $(n+1) \times (n+1)$ matrix

$$A = \begin{pmatrix} I & \underline{a} \\ \underline{a}^T & 0 \end{pmatrix},$$

where $\underline{a} \in \mathbb{R}^n$ and I is $n \times n$ Identity matrix.

- (a) [3] When is A invertible? Give your answer in terms of $\underline{a}$. Justify your answer.
 -  (b) [3] Assuming the condition you found above holds, give an expression for the inverse matrix. *Hint: What is the answer when $n = 1$? Difficult*
- For this problem, A is a 2×2 matrix. Answer true or false, with justification. All of the parts are separate problems. If the statement is false, provide a counterexample. If true, provide proof of the same.
 - (a) [2] If $A^2 = 0$ then A cannot be symmetric nonzero.
 - (b) [2] If A has diagonal entries 1, and $\underline{x}$ is a unit norm vector ($\|\underline{x}\|_2 = 1$) then we can pick $\underline{x}$ such that $\underline{x}^T A \underline{x} = 1$.
 - (c) [2] Similar to above, if A has diagonal entries 1, and $\underline{x}$ is a unit norm vector ($\|\underline{x}\|_2 = 1$) then we cannot pick $\underline{x}$ such that $\underline{x}^T A \underline{x} > 1$.
 - Let A be an $m \times n$ matrix. Let k an integer such that A can be factored as

$$A = BC, \quad \text{where } B \text{ is } m \times k \text{ and } C \text{ is } k \times n. \tag{1}$$

- (a) [3] Show that k has to be at least $\text{rank}(A)$, i.e. $k \geq \text{rank}(A)$.
- (b) [3] When $k = \text{rank}(A)$, how would you compute the decomposition 1 above? How many arithmetic operations would such a computation cost?

- (c) [3] Suppose A is an $m \times n$ matrix with rank k , and a decomposition like in (1) has already been computed (i.e. B and C are known). Now, we would like to compute $A\bar{x}$ for inputs $\bar{x} \in \mathbb{R}^n$. How would you do this computation? How many arithmetic operations would it cost?
6. [8] Consider a robot with unit mass that can move in a frictionless two-dimensional plane. The robot has a constant unit speed in the y -direction (towards north), and it is designed such that we can only apply force in the x -direction. We will apply a (horizontal) force at time t given by f_j for $2j-2 \leq t \leq 2j$ where $j = 1, \dots, n$, so that the applied force is constant over time intervals of length 2. So a force of f_1 is applied for time $0 \leq t \leq 2$, force f_2 for time $2 \leq t \leq 4$ and so on.
- In this problem, we assume that $y_i = i$, the robot is initially at $(0,0)$, and the initial velocity of the robot has no x -component.
- (a) Find the coordinates of the robot at time t , where $0 \leq t \leq 2n$. Your answer should be a function of t and the vector of input forces $\underline{f} \in \mathbb{R}^n$. You do not need the exact expression, but you need to correctly identify the structure of the position function (with variables as the forces applied $\underline{f}$).
- Now, there are $2n$ coins in the plane and the goal is to design a sequence of input forces for the robot to collect the maximum possible number of coins. The robot is designed to collect the i^{th} coin only if it passes through the location of the coin (x_i, y_i) .
- (b) Given the location of the $2n$ coins, describe a method to determine if it is possible for the robot to collect them.
- Hint: With initial position x_0 , initial velocity v_0 and constant acceleration a , the position and velocity at time t are given by $x(t) = x_0 + v_0 t + \frac{1}{2}at^2$, $v(t) = v_0 + at$.
7. [7] What is the nearest neighbor of $x \in \mathbb{R}^n$ among the unit vectors $e_1, \dots, e_n$ in the p -norm sense, for $p = 1, 2, \infty$?

1(a) Solution to $\min \|A\underline{x} - \underline{y}\|_2^2$ is $\underline{x}^* = (A^T A)^{-1} A^T \underline{y}$
 Alternately if $A = QR$ with Q orthogonal,
 then $\underline{x}^* = R^{-1} Q^T \underline{y}$

Students may assume columns of A are linearly independent.

For this problem $\min \|A\underline{x} - B\underline{y}\|^2$, solution is $(A^T A)^{-1} A^T B \underline{y}$
 Alternately it is $R^{-1} Q^T B \underline{y}$

(b) $AB = 0$ or $BA = 0$

(c) From (a), $\underline{x}^* = (A^T A)^{-1} A^T B \underline{y} = 0$ or $R^{-1} Q^T B \underline{y} = 0$,

Since columns of Q (which are linear combinations of columns of A) are perpendicular to columns of B .

Other explanations along the lines of: Any $B\underline{y}$ is perpendicular to any $A\underline{x}$, so the best approximation to $A\underline{x}$ among all possible $B\underline{y}$ is zero.

2 (a) the size of A_n doubles each n , so 2^n .

(b) $A_n = \begin{pmatrix} A_{n-1} & I \\ I & -A_{n-1} \end{pmatrix}$, $A_n^T = \begin{pmatrix} A_{n-1}^T & I \\ I & -A_{n-1}^T \end{pmatrix}$

So if A_{n-1} is symmetric, so is A_n .

Since A_1 is symmetric, it follows that A_n is symmetric by induction.

(c) Similar to above,
 $A_n^2 = \begin{pmatrix} A_{n-1} & I \\ I & -A_{n-1} \end{pmatrix} \begin{pmatrix} A_{n-1} & I \\ I & -A_{n-1} \end{pmatrix} = \begin{pmatrix} A_{n-1}^2 + I & 0 \\ 0 & A_{n-1}^2 + I \end{pmatrix}$

We see that $A_1^2 = I$, so from above $A_2^2 = 2I$, $A_3^2 = 3I$, ...

In general $A_n^2 = nI$.

(d) A_n is a square matrix with

$$A_n \underline{x} = 0 \Rightarrow A_n^2 \underline{x} = 0 \Rightarrow nI \underline{x} = 0 \Rightarrow \underline{x} = 0.$$

So A_n has trivial nullspace & zero nullity. By rank-nullity thm, $\text{rank}(A_n) = \text{no of columns in } A_n = 2^n$.

3 (a) $A = \begin{pmatrix} I & \underline{a} \\ \underline{a}^T & 0 \end{pmatrix}$

check the nullspace of A .

$$A \underline{x} = 0 \Rightarrow \begin{pmatrix} I & \underline{a} \\ \underline{a}^T & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \Rightarrow \underline{x}_1 + \underline{a} x_2 = 0$$

$$\underline{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \begin{matrix} \mathbb{R}^n \\ 1 \times 1 \end{matrix}$$

$\underline{x}_1$ is along $\underline{a}$
 $\downarrow$
 $\underline{a}^T \underline{x}_1 = 0$
 $\underline{x}_1$ is orthogonal to $\underline{a}$

$\downarrow$
 only possible when
 $\underline{a} = 0$ or if $\underline{a} \neq 0$
 then $\underline{x}_1 = 0, x_2 \neq 0$

Thus there are no-nonzero vectors in the nullspace if and only if $\underline{a} \neq 0$.

By rank-nullity A is full rank if & only if $\underline{a} \neq 0$.

(b) For inverse note that when $n=1$, a is scalar & inverse is

$$\begin{pmatrix} 1 & a \\ a & 0 \end{pmatrix}^{-1} = -\frac{1}{a^2} \begin{pmatrix} 0 & -a \\ -a & 1 \end{pmatrix}$$

Consider solving the equations to get $\underline{x}_1, x_2$ in terms of y_1, y_2 .

$$\begin{pmatrix} I & \underline{a} \\ \underline{a}^T & 0 \end{pmatrix} \begin{pmatrix} \underline{x}_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$$\left. \begin{array}{l} \underline{x}_1 + \underline{a} x_2 = y_1 \\ \underline{a}^T \underline{x}_1 = y_2 \end{array} \right\} \Rightarrow \underline{a}^T \underline{x}_1 + \underline{a}^T \underline{a} x_2 = \underline{a}^T y_1$$

$$\underline{a}^T \underline{x}_1 = y_2$$

$$\text{set } x_2 = \frac{\underline{a}^T y_1 - y_2}{\underline{a}^T \underline{a}},$$

Now use this x_2 to obtain x_1 in terms of y_1 & y_2

$$\begin{aligned} \underline{x}_1 &= \underline{y}_1 - \underline{a}x_2 \\ &= \underline{y}_1 - \underline{a} \left(\frac{\underline{a}^T \underline{y}_1 - y_2}{\underline{a}^T \underline{a}} \right) = \left(\underline{I} - \frac{\underline{a} \underline{a}^T}{\underline{a}^T \underline{a}} \right) \underline{y}_1 + \frac{\underline{a}}{\underline{a}^T \underline{a}} y_2 \end{aligned}$$

$$\text{So, } \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \underbrace{\frac{1}{\underline{a}^T \underline{a}} \begin{pmatrix} \underline{a} \underline{a}^T \underline{I} - \underline{a} \underline{a}^T & \underline{a} \\ \underline{a}^T & -1 \end{pmatrix}}_{\text{this is } A^{-1}} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

when a is a scalar, the circled part becomes zero.

4

(a) If A is symmetric, $A^2 = A^T A = 0$

then $\underline{x}^T A^T A \underline{x} = 0$ for every $\underline{x}$

which means $\|A\underline{x}\|_2^2 = 0$ for every $\underline{x}$

so every $\underline{x} \in \mathbb{R}^n$ is in the null space i.e. $A = 0$.
True

(b) $A = \begin{pmatrix} 1 & b \\ c & 1 \end{pmatrix}$ Take $\underline{x} = \underline{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

then $\underline{x}^T A \underline{x} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & b \\ c & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1$.

True

(c) $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ Take $\underline{x} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)^T$

then $\underline{x}^T A \underline{x} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = 2$

False

7 : $p = 2$.

$$\|x - e_i\|_2^2 - \|x\|_2^2 = (x_i - 1)^2 - x_i^2 \\ = -2x_i + 1$$

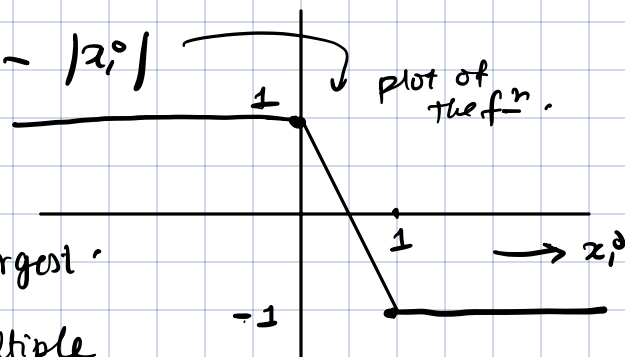
this value is the smallest when x_i^* is the largest.
thus we pick e_i^* such that x_i^* is largest, and declare e_i^* to be closest. If there are multiple i with same largest co-ordinate x_i^* , all the corresponding e_i are nearest.

$p = 1$

$$\|x - e_i\|_1 - \|x\|_1 = |x_i - 1| - |x_i|$$

this is non-increasing, so we
again pick i such that x_i^* is largest.

Similar to above, there could be multiple
nearest neighbours (all i such that $x_i^* > 1$ would result in e_i being a nearest neighbour).



$p = \infty$

$$\|x - e_i\|_\infty - \|x\|_\infty = \max |x_i - e_i| - \max |e_i|$$

For this to be small, we need to pick i such that x_i^* is the largest entry (not largest absolute value).

∴ if $x_i < 0$ there is no point
in replacing x_i^* with $x_i^* - 1$ as this may only increase
the ∞ norm.

we may have multiple nearest neighbours, in particular if
many i have the same largest co-ordinate.

5 (a) Recall that the rows of BC are linear combinations of rows of C . Since C has at most k rows, rows of $A = BC$ are linear combinations of at most k row vectors. So

$$\text{rank}(A) = \text{no of linearly indep rows} \leq k$$

An alternate proof can be given from columns of BC as well.

(b) When $k = \text{rank}(A)$, one way to compute the decomposition above is via QR decomposition (for e.g. by Gram-Schmidt). In this decomposition $B = Q$ is an orthogonal matrix, & $C = R$ is triangular. Cost

$$O(\underbrace{\max\{m, n\}}_{\substack{\text{larger dimension} \\ \text{of matrix}}} k^2) \text{ as discussed in class.}$$

rank

(c) Computing Az directly costs $O(mn)$ as discussed in class.

Instead, we may consider computing $B(Cz)$: compute Cz first then use the result to find $B(Cz)$.

Find Cz costs $k(2n-1)$ operations,

Find $B(Cz)$ costs $m(2k-1)$ operations

Overall $O(k(m+n)) = O(k \max\{m, n\})$ operations.

6 (a) As many of you have realized, the expression is quite complicated, but best expressed recursively as

$$x_{n+1} = x_n + v_n/2 + \frac{f_n(z)^2}{2}$$

x -position at time $2(n+1)$

$$v_{n+1} = v_n + 2f_n$$

x -velocity at time $2(n+1)$

From x_0, v_0 we can obtain $x_1, v_1; x_2, v_2; \dots; x_n, v_n$.

then for $2n \leq t \leq 2n+2$,

$$x_{\text{pos}}(t) = x_n + v_n(t-2n) + \frac{1}{2}a(t-2n)^2$$

$$v(t) = v_n + f_n(t-2n).$$

$x_{\text{pos}}(t)$: x-Position
at time t

the important thing to note is that at a fixed time t , the position (& velocity) are linear functions of applied forces $f_1, f_2, \dots, f_n$.

(b) Let the $2n$ coins be located at
 $(x_1, y_1), (x_2, y_2), \dots, (x_{2n}, y_{2n})$

Note that the robot's y velocity is fixed, so for the robot to collect all coins, its x -position at time $= y_i$ must be x_i .

We must have

$$x_{\text{pos}}(y_1) = x_1$$

$$x_{\text{pos}}(y_2) = x_2$$

⋮

$$x_{\text{pos}}(y_{2n}) = x_{2n}$$

Since (x_i, y_i) are given, there are $2n$ linear equations in n variables $f_1, f_2, \dots, f_n$. If we can solve this system of equations, we obtain the forces to apply for the robot to collect the coins.

Solving equations can be done by row reduction or QR decomposition (the latter finds a least square solution).